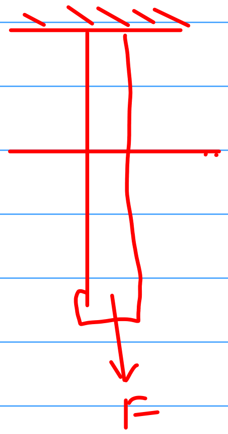
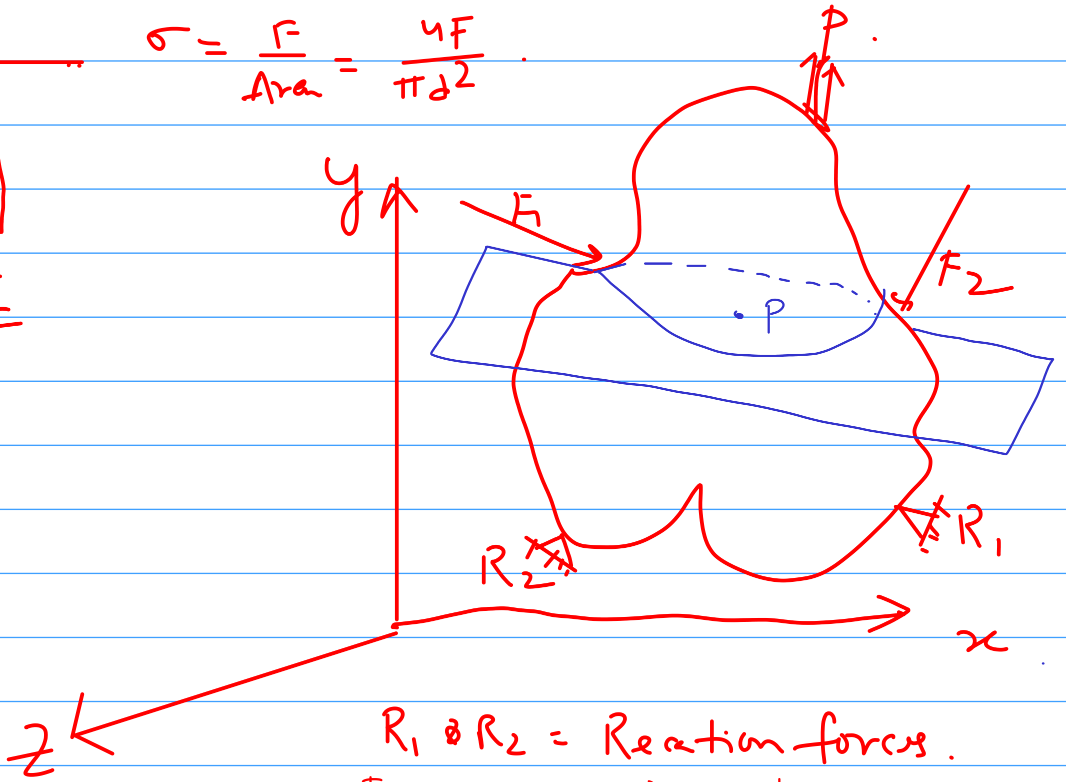
Chapter – 2

Analysis of Stress and Strain

10th Aug 2020



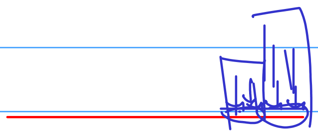
$$\sigma = \frac{F}{Area} = \frac{4F}{\pi d^2}.$$

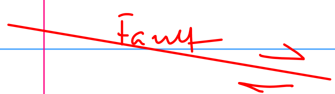


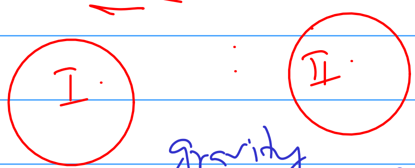
R_1 & R_2 = Reaction forces.

F_1 & F_2 → Point loads.

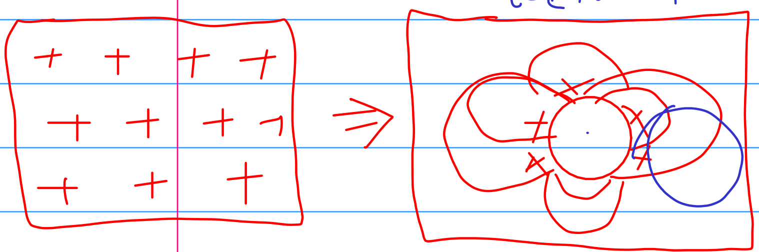
p = → Surface traction.



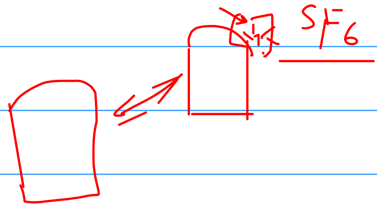




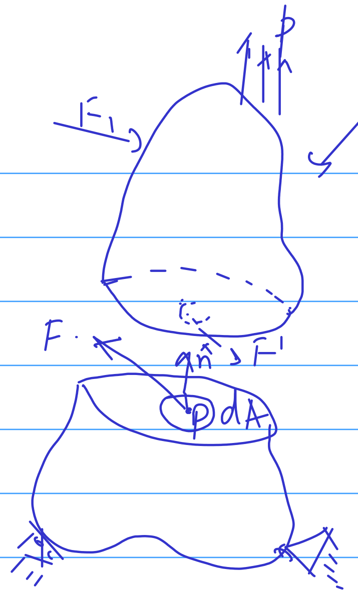
gravity
tectonic forces



fluid pressure



Continuum Mechanics



$\vec{p}_{\text{averaged over a area } \Delta A} = \dfrac{F}{\Delta A}$ — (1)

$\vec{p} = \lim_{\Delta A \to 0} \dfrac{\vec{F}}{\Delta A}$ = Traction at point p.

$$\vec{p}(\vec{x}, \hat{n}) = \lim_{\Delta A \to 0} \frac{F}{\Delta A} \quad \text{— (2)}$$

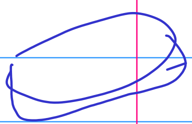
Eqn. (2) indicates

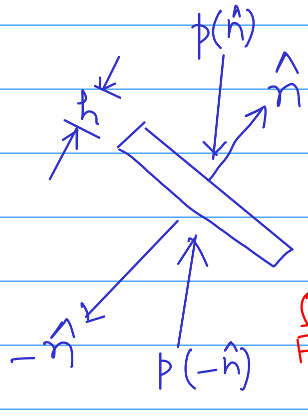
- the traction is a vector.
- this vector is varying from point to point.
- at a given point the traction will be different for different planes.
- $\vec{p}$ is a function of two vectors $\vec{x}, \hat{n}$.

Cauchy. → Concept of "stress"
→ it's a "second order tensor"

1st Law of Cauchy:

(L) $\text{Lt}\ p_{(x,y)} = p'_{(x',y')}$





r = radius of the penny.

Force on face (having $\hat n$) = $\pi r^2 p(\hat n)$

Similarly for face (having $-\hat n$) = $\pi r^2 p(-\hat n)$

~~Traction~~ Force over outer rim = $2\pi r h \hat{t}$.

Force Balance yield

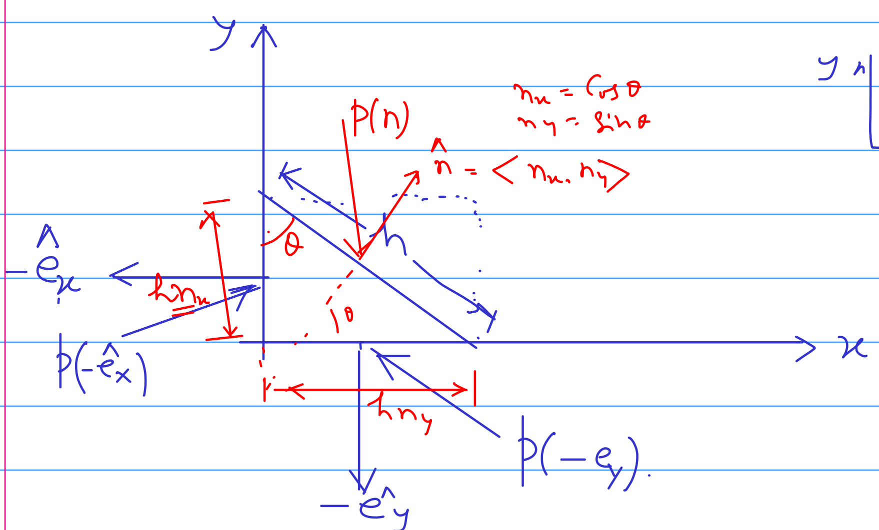
$$\pi r^2 \vec{p}(\hat n) + \pi r^2 \vec{p}(-\hat n) + 2\pi r h \hat{t} = 0 \quad \text{— (3)}$$

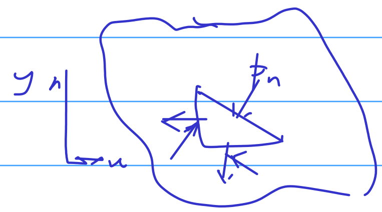
As the thickness 'h' of the slab $\longrightarrow 0$.

$$\pi r^2 p(\hat{n}) + \pi r^2 p(-\hat{n}) = 0$$

$$\boxed{p(-\hat{n}) = -p(\hat{n})} \quad \text{——} \quad (4)$$

Cauchy's 2nd law of Stress.





Let us suppose the element thickness in z-direction = w..

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The of face with outward normal vector $(-\hat{e}_x)$

$= h(n_x)$

Traction on the same face $= p(-\hat{e}_x)$.

The total force acting on this face $= p(-\hat{e}_x)\cdot h \cdot n_x \times w$.

Similarly

The total force acting on the face (with unit normal vector $-\hat{e}_y$)

$= p(-\hat{e}_y)\cdot h \cdot n_y \cdot w$.

For the inclined side.

Force $= p(n) \times h \times w(\hat{n})$

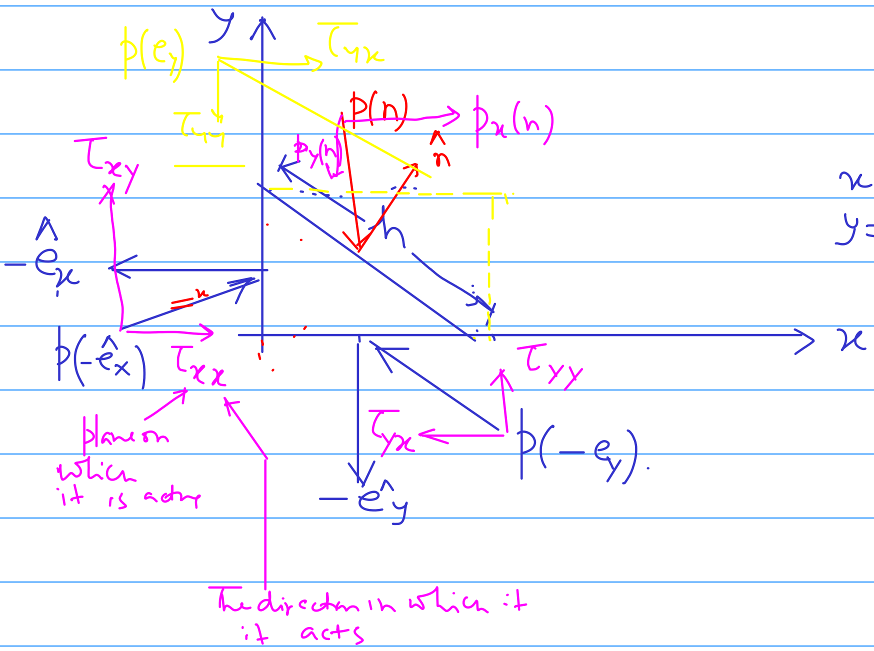
$$hw \cdot n_x \cdot p(-\hat{e}_x) + hwn_y \cdot p(-\hat{e}_y) + hw\, p(\hat{n}) = 0 \quad \text{——} \quad (5)$$

$$\vec{p}(\hat{n}) = -n_x \underline{\vec{p}(-\hat{e}_x)} - n_y \vec{p}(-\hat{e}_y)$$

Cauchy's 1st law $\quad p(-\hat{e}_x) = -p(\hat{e}_x)$

$$\boxed{\vec{p}(n) = n_x\, p(\hat{e}_x) + n_y\, p(\hat{e}_y)} \quad (6)$$

Matrix Representation of Stress in Two dimension



$x = a \Rightarrow x\text{-plane}$

$y = 0 / y = b \Rightarrow$ y-plane.

$\vec{a} = \langle a_i, b_i \rangle$

$$\begin{bmatrix} a_i \\ b_i \end{bmatrix}$$

The of traction vector $p(\hat{e}_x) = \begin{bmatrix} \tau_{xx} \\ \tau_{xy} \end{bmatrix} = \begin{bmatrix} \tau_{xx} & \tau_{xy} \end{bmatrix}^T$

$$p(\hat{e}_y) = \begin{bmatrix} \tau_{yx} \\ \tau_{yy} \end{bmatrix} = \begin{bmatrix} \tau_{yx} & \tau_{yy} \end{bmatrix}^T. \quad (7)$$

Using convention (7) we can re-write Eqn (6) as

$$p(\hat{n}) = n_x \begin{bmatrix} \tau_{xx} \\ \tau_{xy} \end{bmatrix} + n_y \begin{bmatrix} \tau_{yx} \\ \tau_{yy} \end{bmatrix}$$

$$p(\hat{n}) = \begin{bmatrix} \tau_{xx} & \tau_{yx} \\ \tau_{xy} & \tau_{yy} \end{bmatrix} \begin{bmatrix} n_x \\ n_y \end{bmatrix}$$

$$\begin{bmatrix} p_x(n) \\ p_y(n) \end{bmatrix} = \begin{bmatrix} \tau_{xx} & \tau_{yx} \\ \tau_{xy} & \tau_{yy} \end{bmatrix} \begin{bmatrix} n_x \\ n_y \end{bmatrix}$$

$$\vec{p}_x(\hat{n}) = \tau_{xx}\, \hat{n}_x + \tau_{yx}\, \hat{n}_y \quad (9)$$

$$\vec{p}_y(\hat{n}) = \tau_{xy}\, \hat{n}_x + \tau_{yy}\, \hat{n}_y \quad (10)$$

$$\vec{p}(\hat{n}) = \begin{bmatrix} \tau_{xx} & \tau_{xy} \\ \tau_{yx} & \tau_{yy} \end{bmatrix}^T \begin{bmatrix} n_x \\ n_y \end{bmatrix} \quad (11)$$

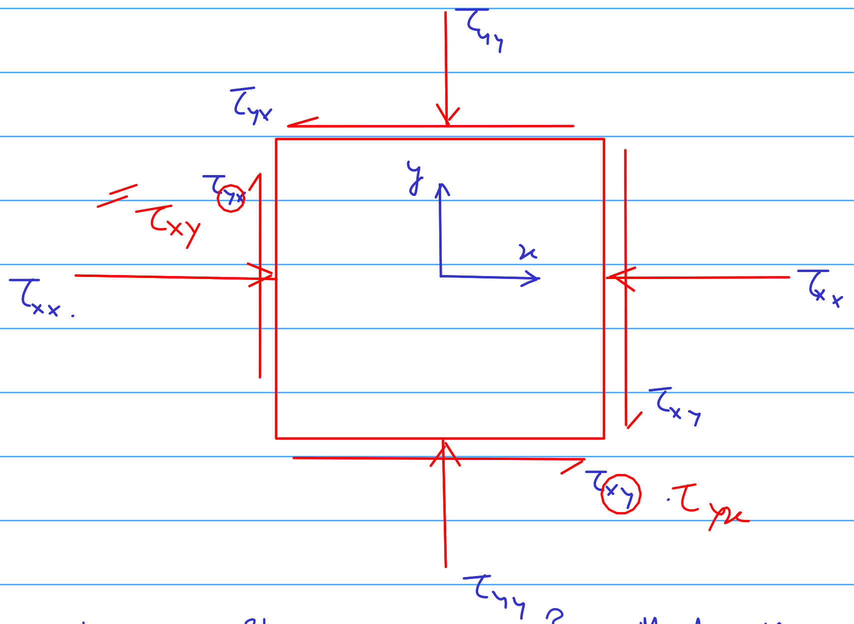
Compact form

$$\boxed{\tilde{p} = \tau^T \hat{n}} \quad (12)$$

$\vec{p}$ $\tilde{p}$ $\tilde{\tilde{q}}$

$\boxed{\tilde{p} = \tau \hat{n}}$ — Symmetry of matrix.

Physical Significance of Stress Tensor



Conventions of Stress used in Rock Mechanics

* Traction components are considered +ve if they are oriented in the direction opposite to the outward unit normal vector.

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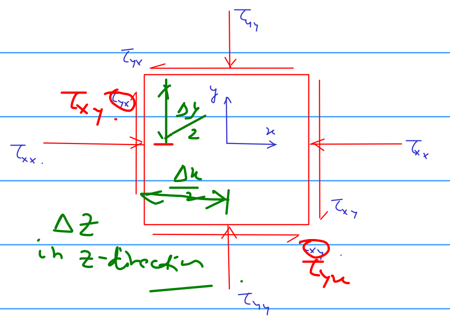
Symmetry of Stress tensor



Consider the traction on Right face of element

Area of face = $\Delta y \cdot \Delta z$.

x-component of force = $\tau_{xx} \, \Delta y \, \Delta z$.

y-component of traction is τ_{xy}.

Net associated with it = $\tau_{xy} \, \Delta y \, \Delta z$.

Moment of this force arm = $\frac{\Delta x}{2}$. (Considering centre as pivot point)

Total moment about z-axis through $(x,y)=(0,0)$

$$= \tau_{xy} \frac{\Delta x \cdot \Delta y \, \Delta z}{2}$$

Similar other moment can be evaluated for the element.
adding all the moments

$$\tau_{xy} \frac{\Delta x \Delta y \Delta z}{2} + \tau_{xy} \frac{\Delta x \Delta y \Delta z}{2} - \tau_{yx} \frac{\Delta x \Delta y \Delta z}{2} - \tau_{yx} \frac{\Delta x \Delta y \Delta z}{2} = 0 \quad (13)$$

$$\boxed{\tau_{xy} = \tau_{yx}} \quad (14)$$

$$\tau_{xz} = \tau_{zx}, \quad \tau_{yz} = \tau_{zy}.$$

$$\tau = \begin{bmatrix} \tau_{xx} & \tau_{yx} \\ \tau_{xy} & \tau_{yy} \end{bmatrix} = \begin{bmatrix} \tau_{xx} & \tau_{xy} \\ \tau_{yx} & \tau_{yy} \end{bmatrix}^T.$$

$$A^T = A$$

$$\boxed{\tau = \tau^T}$$

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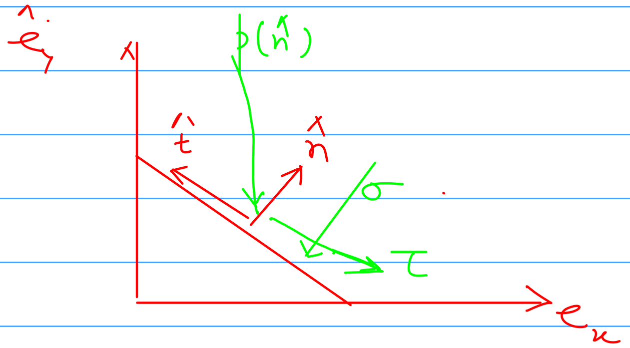
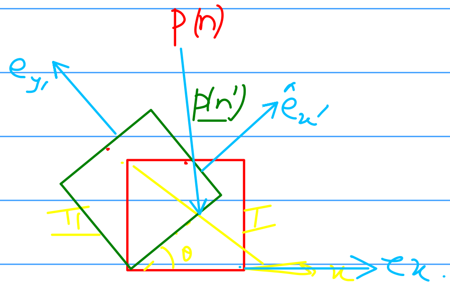
Analysis of Stress in Two Dimensions

Concept of the law of Transformation of Stress Components under a rotation of Co-ordinate System.





$$\boxed{\tau' = L \tau L^T.}$$

$$\hat{n} = \langle n_x, n_y \rangle$$

Unit Vector $\hat{t}$ is $\perp^r$ $\hat{n}$.

$$\hat{t}\cdot\hat{t} = 1.$$

$$\hat{t}_x^2 + t_y^2 = 1$$

Since $\hat{t}$ & $\hat{n}$ are orthogonal

$$\hat{t}\cdot\hat{n} = 0 \Rightarrow t_x n_x + t_y n_y = 0.$$

$$\hat{t} = \pm(\hat{n}_y, -\hat{n}_x)$$

$$\boxed{\hat{t} = (-\hat{n}_y, \hat{n}_x)}$$ — to comply with convention of Rock-Mech

$$\vec{p}_x = \tau_{xx}\hat{n}_x + \tau_{yx}\hat{n}_y \quad \text{— (16)}$$

$$\vec{p}_y = \tau_{xy}\hat{n}_x + \tau_{yy}\hat{n}_y \quad \text{— (17)}.$$

Component of $\vec{p}(n)$ w.r.t $(\hat{n},\hat{t})$ Coordinate System.

$$\left.\begin{array}{l} \sigma = \vec{p}(n)\cdot\hat{n} \\ \tau = \vec{p}(n)\cdot\hat{t}. \end{array}\right\}$$

$$p_n = \begin{bmatrix} p_x \\ p_y \end{bmatrix}\cdot\begin{bmatrix} n_x \\ n_y \end{bmatrix}$$ using 16 & 17

$$= p_x\cdot n_x + p_y n_y.$$

$$= (\tau_{xx}\hat{n}_x + \tau_{yx}\hat{n}_y)\cdot\hat{n}_x + (\tau_{xy}\hat{n}_x + \tau_{yy}\hat{n}_y)\cdot\hat{n}_y.$$

$$= \tau_{xx}n_x^2 + \tau_{yx}n_x\cdot n_y + \tau_{xy}n_x\cdot n_y + \tau_{xx}n_y^2. \quad \text{— (18)}$$

But $\tau_{xy} = \tau_{yx}$

$$\boxed{p_n = \sigma = \tau_{xx}n_x^2 + 2\tau_{xy}n_xn_y + \tau_{yy}n_y^2} \quad \text{— (19)}$$

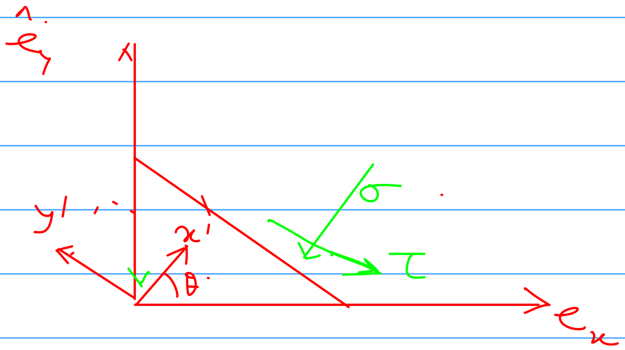
The tangential Component

$$\tau = p_t = \vec{p}(n)\cdot\hat{t}$$

$$p_t = \tau = p(\hat{n}) \cdot \hat{t}$$

$$= \begin{bmatrix} p_x \\ p_y \end{bmatrix} \cdot \begin{bmatrix} -\hat{n}_y \\ \hat{n}_x \end{bmatrix}$$

$$\boxed{p_t = (\tau_{yy} - \tau_{xx}) n_x n_y + \tau_{xx} (n_x^2 - n_y^2)} \quad — (20)$$



$$(\hat{n}, \hat{t}) \approx (x', y')$$

$$p_{x'}(\hat{e}_{x'}) = \tau_{x'x'} = p_n = \tau_{xx} n_x^2 + 2\tau_{xy} n_x n_y + \tau_{yy} n_y^2 \quad — (21)$$

$$p_y(\hat{e}_{y'}) = \tau_{x'y'} = p_t = (\tau_{yy} - \tau_{xx}) n_x n_y + \tau_{xy} (n_x^2 - n_y^2) \quad (22)$$

Similarly

$$p_{y'}(\hat{e}_{y'}) = \tau_{y'y'} = \tau_{xx} n_y^2 - 2\tau_{xy} n_x n_y + \tau_{yy} n_x^2 \quad — (23)$$

$$p_{x'}(\hat{e}_{y'}) = \tau_{y'x'} = (\tau_{yy} - \tau_{xx}) n_x n_y + \tau_{xy} (n_x^2 - n_y^2) \quad (24)$$

Alternative Common Notation of Stress Transformation

$$\langle n_x \; n_y \rangle = \langle \cos\theta, \sin\theta)$$

$$\tau_{x'x'} = \tau_{xx} \cos^2\theta + 2\tau_{xy} \sin\theta \cdot \cos\theta + \tau_{yy} \sin^2\theta \quad — (25)$$

$$\tau_{y'y'} = \tau_{xx} \sin^2\theta - 2\tau_{xy} \sin\theta \cdot \cos\theta + \tau_{yy} \cos^2\theta \quad — (26)$$

$$\tau_{x'y'} = (\tau_{yy} - \tau_{xx}) \sin\theta \cdot \cos\theta + \tau_{xy} (\cos^2\theta - \sin^2\theta) \quad — (27)$$

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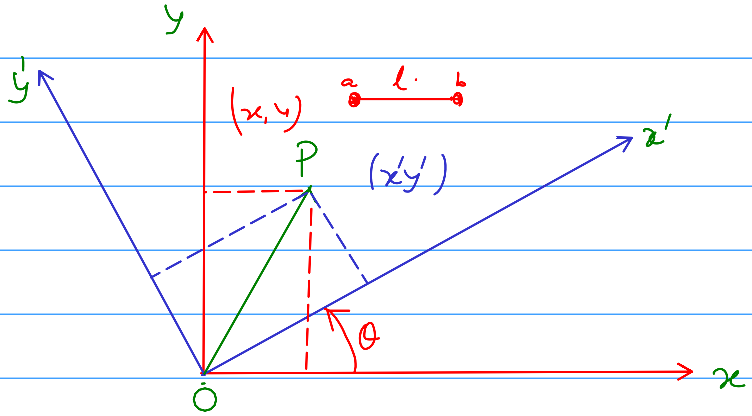


$$(x\ y) \longrightarrow (x', y')$$

$$x' = x\cos\theta + y\sin\theta$$
$$y' = -x\sin\theta + y\cos\theta\,.$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}}_{L} \begin{bmatrix} x \\ y \end{bmatrix} \quad \text{——— } \underline{\underline{27\ (a)}}$$

$\vec{OP'} = L\,\vec{OP}$.

$$L = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \quad \text{————— } \underline{(28)}$$

$$\begin{bmatrix} \tau_{x'x'} & \tau_{x'y'} \\ \tau_{y'x'} & \tau_{y'y'} \end{bmatrix} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \begin{bmatrix} \tau_{xx} & \tau_{xy} \\ \tau_{yx} & \tau_{yy} \end{bmatrix} \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \quad \text{——— } \underline{(29)}$$

$$\boxed{\tau' = L\,\tau\,L^T} \text{ —— } 30$$

$$\boxed{\tilde{p}' = L\,\tilde{p}}$$

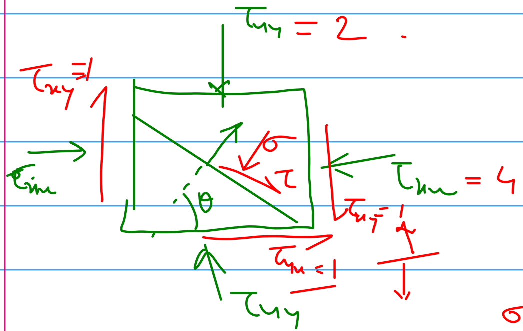
Physical based View point of Stress Transformation.

Using $\rightarrow \begin{cases} \cos^2\theta - \sin^2\theta = \cos 2\theta \\ 2\sin\theta\cdot\cos\theta = \sin 2\theta . \end{cases}$

Eqn (19) & (20) can be simplified to

$$\sigma = \frac{1}{2}(\tau_{xx} + \tau_{yy}) + \frac{1}{2}(\tau_{xx} - \tau_{yy})\cos 2\theta + \tau_{xy}\sin 2\theta \quad \text{———} \quad (31)$$

$$\tau = \frac{1}{2}(\tau_{yy} - \tau_{xx})\sin 2\theta + \tau_{xy}\cos 2\theta . \quad \text{———} \quad (32)$$



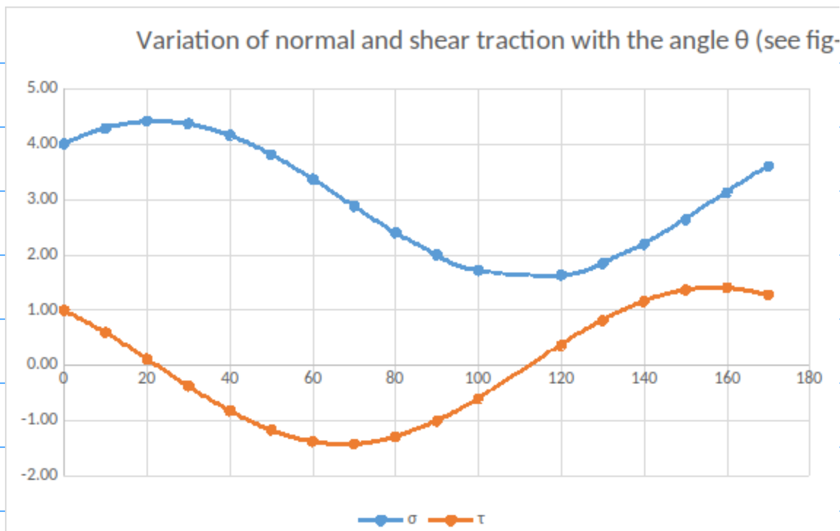
$\tau_{xx} = 4$ MPa

$\tau_{yy} = 2$ MPa.

$\tau_{xy} = 1$ MPa.

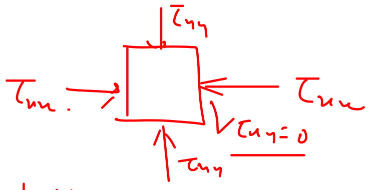
$\sigma, \tau \Rightarrow 0 < \theta < 180^\circ$

θ	σ	τ
0	4.00	1.00
10	4.28	0.60
20	4.41	0.12
30	4.37	-0.37
40	4.16	-0.81
50	3.81	-1.16
60	3.37	-1.37
70	2.88	-1.41
80	2.40	-1.28
90	2.00	-1.00
100	1.72	-0.60
120	1.63	0.37
130	1.84	0.81
140	2.19	1.16
150	2.63	1.37
160	3.12	1.41
170	3.60	1.28



From Eqn (32).

$$\tan 2\theta = \frac{2\tau_{xy}}{(\tau_{xx} - \tau_{yy})} \quad \text{———} \quad (33)$$



Note: $\tau_{xy} = 0$ then the plane with $\hat{n} = e_x$ is already a shear free plane

So Eqn (33) gives $\theta = 0°$, $2\theta = 180°$ $\therefore \theta = \frac{\pi}{2}$.

In general. What ever be the value of $\tau_{xx}, \tau_{xy}, \tau_{yy}$. $\exists$ always two roots for Eqn (33) differing by a value π in the range $0 < 2\theta < 2\pi$ root diff by π

$0 < \theta < \pi$ $\longrightarrow$ Do — by $\frac{\pi}{2}$.

$$\tan 2\theta = \frac{2\tau_{xy}}{\tau_{xx} - \tau_{yy}} = x.$$

$$\therefore \sin 2\theta = \pm \frac{x}{\sqrt{1+x^2}} = \tau_{xy}\left[\tau_{xy}^2 + \frac{1}{4}(\tau_{xx} - \tau_{yy})^2\right]^{-\frac{1}{2}} \quad (34)$$

$$\cos 2\theta = \frac{1}{2}(\tau_{xx} - \tau_{yy})\left[\frac{1}{4}(\tau_{xx} - \tau_{yy})^2 + \tau_{xy}^2\right]^{-\frac{1}{2}} \quad (35)$$

using these values of $\sin 2\theta$ & $\cos 2\theta$ in Eqn (31) we get.

$$\sigma = \frac{1}{2}(\tau_{xx} + \tau_{yy}) \pm \frac{\frac{1}{2} \times \frac{1}{2}(\tau_{xx} - \tau_{yy})^2}{\sqrt{\tau_{xy}^2 + \frac{1}{4}(\tau_{xx} - \tau_{yy})^2}} \pm \frac{\tau_{xy}^2}{\sqrt{\tau_{xy}^2 + \frac{1}{4}(\tau_{xx} - \tau_{yy})^2}}$$

$$\sigma = \frac{1}{2}(\tau_{xx} + \tau_{yy}) \pm \sqrt{\tau_{xy}^2 + \frac{1}{4}(\tau_{xx} - \tau_{yy})^2} \quad (36)$$

$$\sigma_1 = \frac{1}{2}(\tau_{xx} + \tau_{yy}) + \sqrt{\tau_{xy}^2 + \frac{1}{4}(\tau_{xx} - \tau_{yy})^2} \quad (37)$$

$$\sigma_2 = \frac{1}{2}(\tau_{xx} + \tau_{yy}) - \sqrt{\tau_{xy}^2 + \frac{1}{4}(\tau_{xx} - \tau_{yy})^2} \quad (38)$$

63 present at end of class

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Diff. Eqn (31)

$$\frac{d\sigma}{d\theta} = -(\tau_{xx}-\tau_{yy})\sin 2\theta + 2\tau_{xy}\cos 2\theta .$$

$$\Rightarrow = (\tau_{yy}-\tau_{xx})\sin 2\theta + 2\tau_{xy}\cos 2\theta .$$

$$= 2\left[\frac{1}{2}(\tau_{yy}-\tau_{xx})\sin 2\theta + \tau_{xy}\cos 2\theta\right] . \quad \text{Eqn } (32)$$

$$= 2\tau .$$

$$\frac{d^2\sigma}{d\theta} = 2(\tau_{yy}-\tau_{xx})\cos 2\theta - 4\tau_{xy}\sin 2\theta .$$

$$= -2(\tau_{xx}-\tau_{yy})\cos 2\theta - 4\tau_{xy}\sin 2\theta .$$

Case-I

$\tau_{xx} > \tau_{yy}$ & $\tau_{xy} > 0$

$\frac{d^2\sigma}{d\theta} < 0 .$ [Max value of σ $\forall\ \theta = ?$]

$\tan 2\theta > 0$

$0 < 2\theta_1 < \frac{\pi}{2} .$

$0 < \theta_1 < \frac{\pi}{4} .$

$\forall\ \tau_{xx} > \tau_{yy}$ & $\tau_{xy} > 0 \Rightarrow 0 < \theta_1 < \pi/4 .$ —— (41)

Case-II

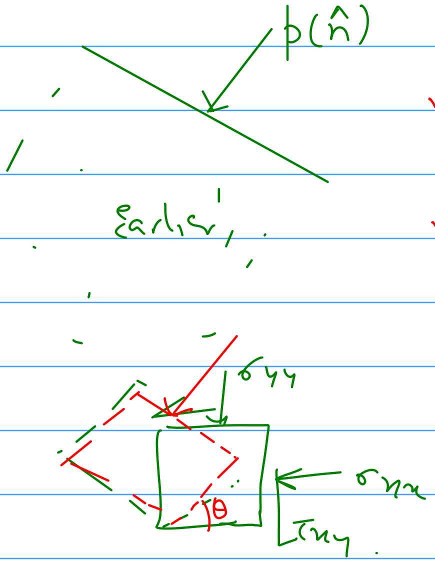
$\tau_{xx} < \tau_{yy}$ & $\tau_{xy} > 0 \Rightarrow \frac{\pi}{4} < \theta_1 < \frac{\pi}{2}$ —— (42)

Case-iii

$\tau_{xx} < \tau_{yy}$ & $\tau_{xy} < 0$ $\quad \frac{\pi}{2} < \theta_1 < \frac{3\pi}{4}$ —— (43)

Case-IV $\tau_{xx} > \tau_{yy}$ & $\tau_{xy} < 0 \Rightarrow \frac{3\pi}{4} < \theta_1 < \pi .$ —— (44)

Alternative Method for finding Principal Stresses and Principal Directions (Matrix Method).



✓ $\tilde{p}(\hat{n}) = \underline{\underline{\sigma}}\,\hat{n}$ [$\underline{\sigma \text{ is yet to Known}}$]

✓ $\tilde{p}(\hat{n}) = \tau^T \hat{n} = \tau\,\hat{n}$

$$\boxed{\sigma\,\hat{n} = \tau\,\hat{n}} \qquad \text{—— (45)}$$

σ: Scalar; τ: $\underline{\text{Matrix}}$.

I = Identity $\underline{\text{Matrix}}$.

$$\tau\hat{n} = \sigma I \hat{n}$$

$$(\tau - \sigma I)\hat{n} = 0 \qquad \text{—— } \underline{(46)}$$

← Std Eigen Value & Eigen Vector $\underline{\text{problem}}$

$$\left\{\begin{bmatrix} \tau_{xx} & \tau_{yx} \\ \tau_{xy} & \tau_{yy} \end{bmatrix} - \sigma \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right\}\begin{Bmatrix} \hat{n}_x \\ \hat{n}_y \end{Bmatrix} = 0$$

$$\begin{bmatrix} (\tau_{xx}-\sigma) & \tau_{xy} \\ \tau_{xy} & (\tau_{yy}-\sigma) \end{bmatrix}\begin{bmatrix} n_x \\ n_y \end{bmatrix} = 0$$

$$(\tau_{xx}-\sigma)n_x + \tau_{xy}\,n_y = 0 \qquad \text{—— (47)}$$

$$\tau_{xy}\hat{n}_x + (\tau_{yy}-\sigma)n_y = 0 \qquad \text{—— (48)}$$

Using Gauss Elimination Steps 49 ~~58~~

$$\left[\sigma^2 - (\tau_{xx}+\tau_{yy})\sigma + (\tau_{xx}\tau_{yy} - \tau_{xy}^2)\right] n_y^n = 0 \quad \text{(51)}$$

$$\sigma^2 - \underbrace{(\tau_{xx}+\tau_{yy})}\sigma + \underbrace{(\tau_{xx}\tau_{yy} - \tau_{xy}^2)} = 0 \quad \text{(52)}$$

$\det |(\tau - \sigma I)|$

Invariants

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Roots of Eqn (52) if they are σ_1 & σ_2.

Substituting in (47) & (48) one eqn will be independent of other:

in that case using Eqn (47) & the roots σ_1 & σ_2.

$$\tan\theta = \frac{n_y}{n_x} = \frac{2\tau_{xy}}{(\tau_{xx}-\tau_{yy}) \pm \sqrt{4\tau_{xy}^2 + (\tau_{xx}-\tau_{yy})^2}}$$

after few steps of rationalisation

$+ve \Rightarrow \sigma_1$
$-ve \rightarrow \sigma_2$

Graphical Representation of Stress in 2D

Mohr's Circle.

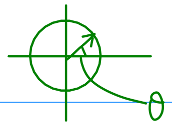
In (31) & (32) using

$$\tau_{xx} = \sigma_1, \quad \tau_{yy} = \sigma_2 \; \& \; \tau_{xy} = 0.$$

Reduces to

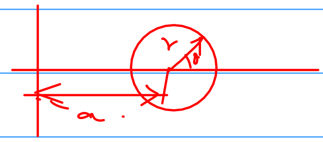
$$\sigma = \frac{(\sigma_1+\sigma_2)}{2} + \left(\frac{\sigma_1-\sigma_2}{2}\right)\cos 2\theta \quad \text{(54)}$$

$$\tau = -\left(\frac{\sigma_1-\sigma_2}{2}\right)\sin 2\theta \quad \text{(55)}$$



$x = r\cos\theta$
$y = r\sin\theta$. parametric eqn of Circle

$x = a + r\cos\theta$.
$y = r\sin\theta$.



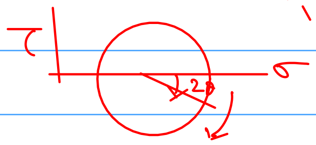
Eqn (54) & (55) represent the parametric Eqn of Circle in (σ, τ) plane

Centre is a point $\left(\frac{\sigma_1+\sigma_2}{2}\right)$ & $\tau = 0$.

$$\text{Radius} = \left(\frac{\sigma_1-\sigma_2}{2}\right).$$

$\theta \rightarrow (-\theta)$.

Instead of θ moving in anticlockwise direction it moves in clockwise direction.



Problem is

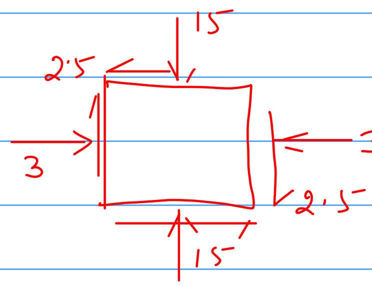
The State of stress at a point is given as under.

$$\tau = \begin{bmatrix} 3 & 2.5 \\ 2.5 & 15 \end{bmatrix}$$

Determine the Value of the principal Stress and the direction of the principal stress. & also Verify the Solution using Mohr's Circle.

Soln:

$$(\tau - \sigma I)\hat{n} = 0$$



$$|\tau - \sigma I| = 0$$

$$\begin{vmatrix} 3-\sigma & 2.5 \\ 2.5 & 15-\sigma \end{vmatrix} = 0$$

$$(3-\sigma)(15-\sigma) - 2.5^2 = 0$$

Blank Page

or $\sigma^2 - 18\sigma + 45 - 2.5^2 = 0$

or, $\sigma^2 - 18\sigma + 38.75 = 0$.

$$\left.\begin{aligned} \sigma_1 &= 15.5 \\ \sigma_2 &= 2.5 \end{aligned}\right\}$$

Now using the major principal stress $\left(\overset{\text{value}}{15.5}\right)$ in the matrix Eqn.

$$\begin{bmatrix} 3-15.5 & 2.5 \\ 2.5 & 15-15.5 \end{bmatrix}\begin{bmatrix} \hat{n}_1 \\ \hat{n}_2 \end{bmatrix} = 0$$

$$\begin{pmatrix} -12.5 & 2.5 \\ 2.5 & -0.5 \end{pmatrix}\begin{pmatrix} n_1 \\ n_2 \end{pmatrix} = 0$$

$$\left(\begin{array}{cc|c} -12.5 & 2.5 & 0 \\ 2.5 & -0.5 & 0 \end{array}\right) \rightarrow \quad R_1 \rightarrow \frac{R_1}{2.5}$$

$$R_2 \rightarrow \frac{R_2}{0.5}$$

$$\left(\begin{array}{cc|c} -5 & 1 & 0 \\ 5 & -1 & 0 \end{array}\right) \qquad R_2 \rightarrow R_2(-1).$$

$$\left(\begin{array}{cc|c} -5 & 1 & 0 \\ -5 & 1 & 0 \end{array}\right) \qquad R_2 \rightarrow R_1 + R_2.$$

$$\left(\begin{array}{cc|c} -5 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$$\Rightarrow \quad -5n_1 + n_2 = 0.$$

or, $n_1 = \dfrac{n_2}{5}$ let $n_2 = k$.

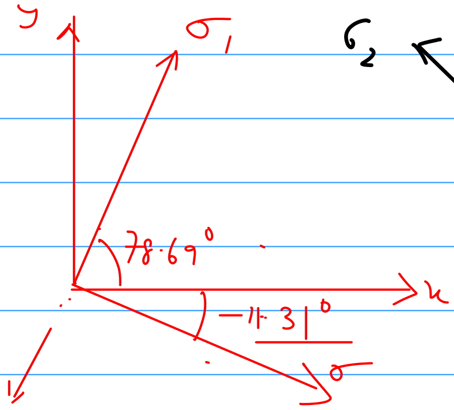
$\Rightarrow n_1 = \dfrac{k}{5}$.

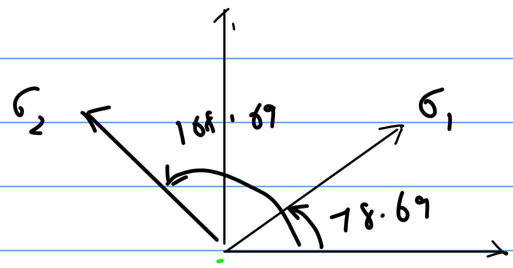
Since $n_1^2 + n_2^2 = 1$

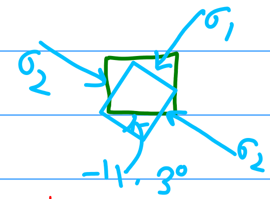
$$\left(\frac{k}{5}\right)^2 + k^2 = 1 \Rightarrow k^2 = \frac{5^2}{1+5^2}.$$

σ_1 $$k = \frac{5}{\sqrt{1+5^2}} \Rightarrow n_1 = \frac{1}{\sqrt{26}}$$

For the σ_1 plane it's normal will make an angle of $\cos^{-1}\left(\frac{1}{\sqrt{26}}\right) \rightarrow \underline{78.69^\circ}$ with x-axis.







Now using the minor principal stress $\sigma_2 = 2.5$

Direction of / eigen vector of σ_2.

$$\begin{pmatrix} 3-2.5 & 2.5 \\ 2.5 & 15-2.5 \end{pmatrix} \begin{pmatrix} \hat{n}_1 \\ \hat{n}_2 \end{pmatrix} = 0$$

$$\begin{pmatrix} 1 & 5 \\ 1 & 5 \end{pmatrix} \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} = 0$$

let $n_2 = k$. $n_1 = -5k$

$$n_1 = -5n_2.$$

<u>Normalizing</u>

$$k^2 + 25k^2 = 1$$

$$k = \frac{1}{\sqrt{26}}, \quad n_1 = \frac{-5}{\sqrt{26}} \Big\}$$

This shows that for the σ_2 plane, its normal will make an angle of $\cos^{-1}\left(\frac{-5}{\sqrt{26}}\right) = -\cancel{11.31}.\ 168.69$

Attendance (42)

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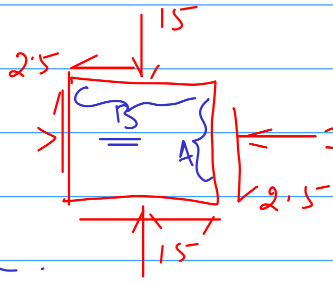
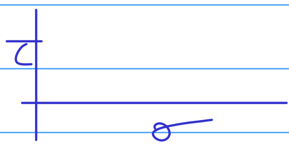


Step-1 :- 

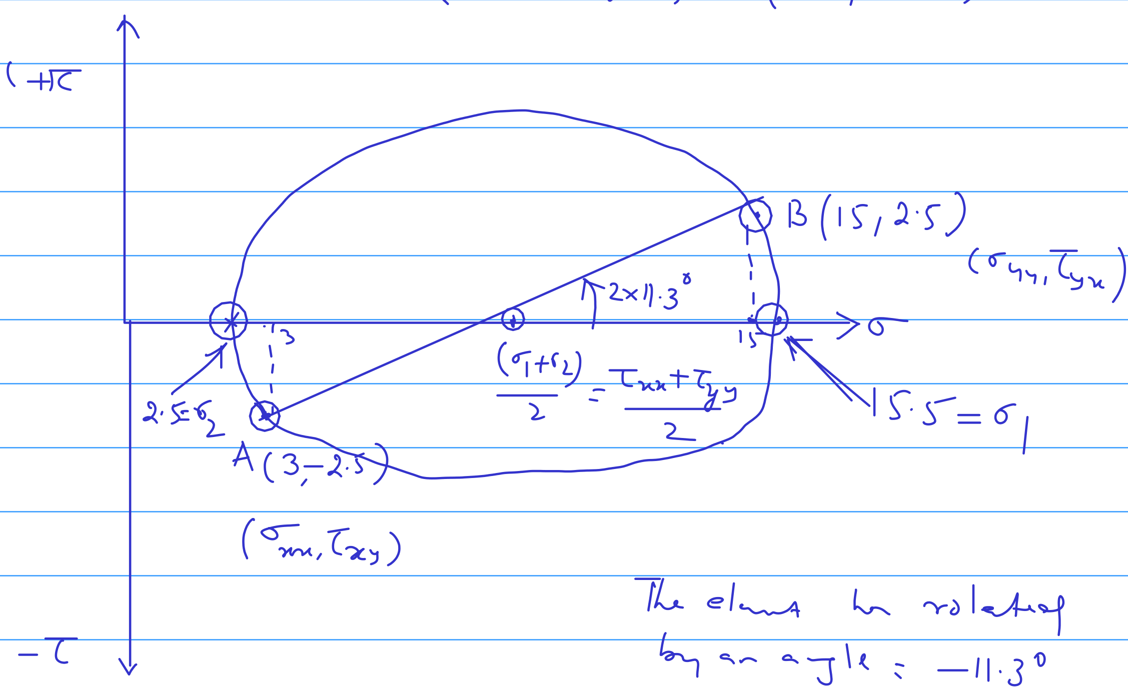
Both axes must have Same Scale.

Step-2 first Concn.. on x-plane on Right
Then " y-plane on top.

Shear Stress is (+ve) if it tends to rotate rock element Anticlock
(-ve) —— also —— Colockwise

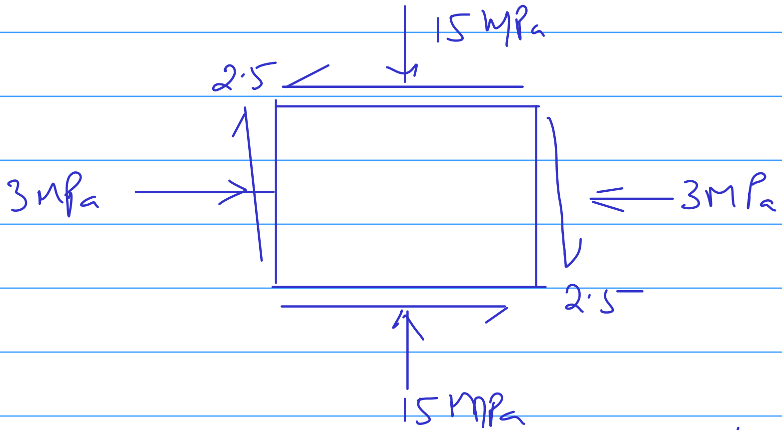
Step-3 plot the points. A & B on M.C

$A = (3, -2.5)$, $B(15, +2.5)$

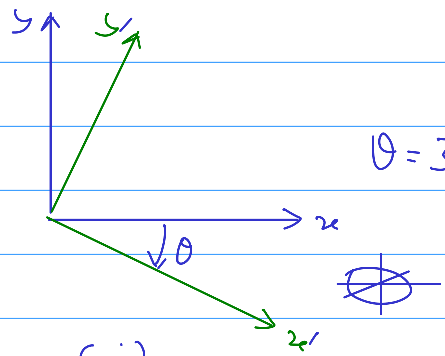


The element has rotated by an angle = -11.3^0

Problem:- The vertical and horizontal stress at the middle of a Coal pillar are found to be 15 MPa and 3 MPa respectively. The Shear stress is 2.5 MPa as shown in the figure below.

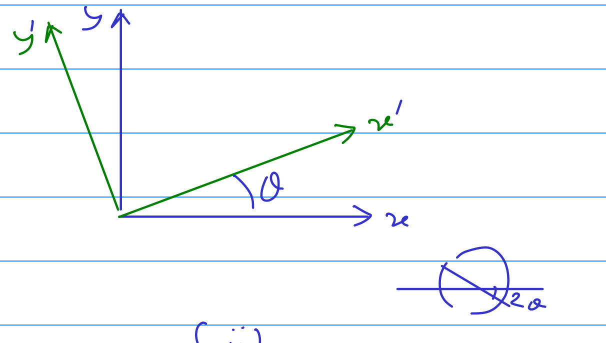


Determine the stresses in the $x'y'$ cordinate system as shown in figures (i) & (ii) below.
Then Verify the stress using Mohr's circle.



(i)

$\theta = 30°$



(ii)

Ans-1 :- (i) $\tau_{x'x'} = 3.834$ (ii) 8.1651

$\tau_{y'y'} = 14.164$ $\quad 9.8349$

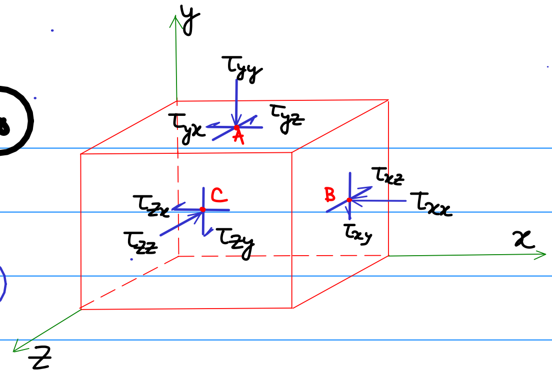
$\tau_{x'y'} = -3.946$ $\quad 6.4462$

Stresses in 3-D

$$\hat{n} = (\hat{n}_x, \hat{n}_y, \hat{n}_z)$$

$$n_x^2 + n_y^2 + n_z^2 = 1$$

$$\vec{p}(\hat{n}) = n_x \vec{p}(\hat{e}_x) + n_y \vec{p}(\hat{e}_y) + n_z \vec{p}(e_z)$$



$$p(\hat{e}_x) = [\tau_{xx},\ \tau_{xy}\ \ \tau_{xz}]^T \quad \text{—} \quad (60)$$

$$p(\hat{e}_y) = [\tau_{yx}\ \ \tau_{yy}\ \ \tau_{yz}]^T \quad \text{—} \quad (61)$$

$$p(\hat{e}_z) = [\tau_{zx}\ \ \tau_{zy}\ \ \tau_{zz}]^T \longrightarrow (62)$$

Substitute (60) to (62) into (59)

$$p_x(\hat{n}) = \tau_{xx} n_x + \tau_{yx} n_y + \tau_{zx} n_z. \quad \text{—} \quad (63)$$

$$p_y(\hat{n}) = \tau_{xy} n_x + \tau_{yy} n_y + \tau_{zy} n_z \quad \text{—} \quad (64)$$

$$p_z(\hat{n}) = \tau_{xz} n_x + \tau_{yz} n_y + \tau_{zz} n_z \quad \text{—} \quad (65)$$

$$\tilde{p} = \tau^T \hat{n}$$

$$\begin{bmatrix} p_x(n) \\ p_y(n) \\ p_z(n \end{bmatrix} = \begin{bmatrix} \tau_{xx} & \tau_{yx} & \tau_{zx} \\ \tau_{xy} & \tau_{yy} & \tau_{zy} \\ \tau_{xz} & \tau_{yz} & \tau_{zz} \end{bmatrix} \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} \quad \text{—}$$

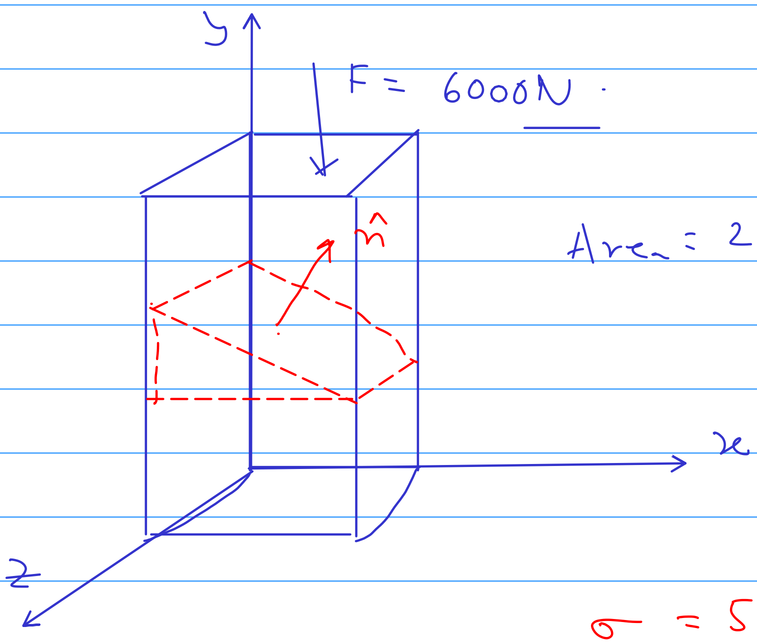
$$\begin{bmatrix} p_x(n) \\ p_y(n) \\ p_z(n \end{bmatrix} = \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix}^T \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} \quad \text{—} \quad (66)$$

$$\tau_{xy} = \tau_{yx}, \quad \tau_{yz} = \tau_{zy}, \quad \tau_{zx} = \tau_{xz} \quad \text{—} \quad (67)$$

$$\boxed{\tilde{p}(\hat{n}) = \tau \hat{n}}$$

Ques: A rectangular rock of cross section $2\,cm \times 3\,cm$ is subjected to a compressive force of 6000 N. If the axes are as shown in figure below, determine the normal and shear stresses on the plane whose normal has the following dc's.

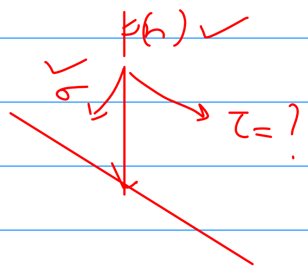
(i) $n_x = n_y = \frac{1}{\sqrt{2}}$ & $n_z = 0$



Area $= 2 \times 3\ cm^2 = 6\ cm^2$

$\sigma = 500\ N/cm^2$.

$p(n) = p_y(n) = n_y \tau_{yy} = 500\ N/cm^2$.



$$|p(n)|^2 = |\sigma|^2 + |\tau|^2$$

$$\tau = 500\ N/cm^2$$

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Solution to the above problem.

normal to the plane is given in 3D as

$$\hat{n} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + 0\hat{k}.$$

The Stress tensor is to be considered in 3D.
We therefore are interested to know the traction on the given plane.
The traction on any plane is given by

$$\tilde{p}(\hat{n}) = \tau^T\hat{n} = \tau\hat{n} \quad \text{— (1)}$$

From the given Configuration

$$\tau = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \tau_{yy} & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \& \quad \tau_{yy} = \frac{6000\ N}{6\ cm^2} = 1000\ N/cm^2 \quad \text{— (2)}$$

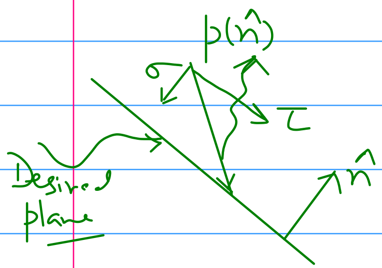
expanding (1) we get —

$$\left.\begin{aligned} p_x(n) &= \tau_{xx} n_x + \tau_{yx} n_y + \tau_{zx} n_z \\ p_y(n) &= \tau_{xy} n_x + \tau_{yy} n_y + \tau_{zy} n_y \\ p_z(n) &= \tau_{xz} n_x + \tau_{yz} n_y + \tau_{zz} n_z. \end{aligned}\right\}$$

Since all components of Stress matrix are zero EXCEPT τ_{yy}
So 2nd Eqn prevails

$$\therefore \quad p_y(n) = \tau_{yy}\hat{n}_y = 1000 \times \frac{1}{\sqrt{2}}$$

$$\& \quad p_x(n) = p_z(n) = 0. \qquad \Rightarrow \quad p(\hat{n}) = \frac{1000}{\sqrt{2}}\hat{j}$$



On the desired plane σ can be found as the dot product of $p(\hat{n})$ and $\hat{n}$

or, $\sigma = p(\hat{n}) \cdot \hat{n}$

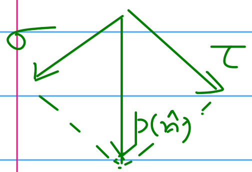
$$= \left(\frac{1000}{\sqrt{2}}\hat{j}\right)\cdot\left(\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}\right)$$

$$= \frac{1000}{2} = 500\ N/cm^2.$$

Also,

$$|p(n)|^2 = |p_x(n)|^2 + |p_y(n)|^2 + |p_z(n)|^2$$

$$\therefore \quad |p(n)|^2 = |p_y(n)|^2 = \left(\frac{1000}{\sqrt{2}}\right)^2$$



$$\Rightarrow \quad \tau^2 = |p(n)|^2 - |\sigma|^2$$

$$= \left(\frac{1000}{\sqrt{2}}\right)^2 - (500)^2$$

$$\tau^2 = 250{,}000\ (N/cm^2)^2$$

or $\boxed{\tau = 500\ N/cm^2}$

Solution given after the regular class

28/08/2020

Principal Stress and Principal Directions in 3D

$$\vec{p}(\hat{n}) = \sigma \hat{n}.$$

σ = some scalar.

But $\vec{P}(\hat{n}) = \tau \hat{n}$.

So we can say

$$\tau \hat{n} = \sigma \hat{n}.$$

$$\tau \hat{n} = \sigma \underline{I} \hat{n}$$

$$(\tau - \sigma \underline{I})\hat{n} = 0. \quad \leftarrow \text{Eigen value problem.}$$

$$\begin{bmatrix} \tau_{xx}-\sigma & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy}-\sigma & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz}-\sigma \end{bmatrix} \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} = 0 \quad \text{—— (68)}$$

$\langle n_x, n_y, n_z \rangle = \langle 0, 0, 0 \rangle$ obvious soln.

Not admissible $\rightarrow$ $\underline{\hat{n} \cdot \hat{n} = 1}$

Admissible Soln can only be found is

$$\det[\tau - \sigma \underline{I}] = 0$$

Expansion

$$\sigma^3 - I_1 \sigma^2 - I_2 \sigma - I_3 = 0 \quad \text{—— (69)}$$

where

$$I_1 = \tau_{xx} + \tau_{yy} + \tau_{zz} \quad \text{—— (70)}$$

$$I_2 = \tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2 - \tau_{xx}\tau_{yy} - \tau_{xx}\tau_{zz} - \tau_{yy}\tau_{zz} . \quad (71)$$

$$I_3 = \tau_{xx}\tau_{yy}\tau_{zz} + 2\tau_{xy}\tau_{xz}\tau_{yz} - \tau_{xx}\tau_{yz}^2 - \tau_{yy}\tau_{xz}^2 - \tau_{zz}\tau_{xy}^2 \quad (72)$$

eqn 69 $\longrightarrow$ Real Roots (3-roots).

$$\sigma_1 \geq \sigma_2 \geq \sigma_3 .$$

$\hat{n}^{(1)}, \hat{n}^{(2)}, \hat{n}^{(3)} \longrightarrow$ 3-Eigen Vectors.

$$\hat{n}^{(1)} = \langle \hat{n}^{(1)}_{xx} \quad \hat{n}^{(1)}_{yy} \quad n^{(1)}_{zz} \rangle$$

Orthogonality of Principal Directions

Little Concepts.

$$\vec{u}\cdot\vec{v} = [\vec{u}]^T [v]$$

$$= \begin{bmatrix} u_x & u_y & u_z \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$$

$$= u_x v_x + u_y v_y + u_z v_z .$$

$$= \vec{u}\cdot\vec{v} .$$

Transpose opn of Matrix

$$\left.\begin{aligned} (AB)^T &= B^T A^T . \\ (A^T)^T &= A . \end{aligned}\right\} \quad \text{—— (73)}$$

Consider $\sigma_1 \neq \sigma_2$.

principal directions $\hat{n}^{(1)}, \hat{n}^{(2)}$ — (1) & (2) super scripts

$$\left.\begin{aligned} \tau \hat{n}^{(1)} &= \sigma_1 \hat{n}^{(1)} \\ \tau \hat{n}^{(2)} &= \sigma_2 n^{(2)} \end{aligned}\right\}$$

Taking inner product on both sides by $\hat{n}^{(2)}$ for 1st Eqn

$$[n^{(2)}]^T [\tau n^{(1)}] = [n^{(2)}] [\sigma_1 \hat{n}^{(1)}]$$
$$= \sigma_1 [n^{(2)}]^T \hat{n}^{(1)} \quad \text{—— (74)}$$

Again for 2nd Eqn Both side inner product $\hat{n}^{(1)}$

$$[n^{(1)}]^T \tau n^{(2)} = [n^{(1)}]^T \sigma_2 n^{(2)}$$
$$= \sigma_2 [n^{(1)}]^T n^{(2)}. \quad \text{—— (75)}$$

Taking transpose on both sides of (75)

$$\left\{[n^{(1)}]^T \tau n^{(2)}\right\}^T = \left\{\sigma_2 [n^{(1)}]^T n^{(2)}\right\}^T \quad \text{—— (76)}$$

$$[n^{(2)}]^T \tau^T n^{(1)} = \sigma_2 [n^{(2)}]^T n^{(1)}$$

But $\tau^T = \tau$

$$[n^{(2)}]^T \tau n^{(1)} = \sigma_2 [n^{(2)}]^T n^{(1)} \quad \text{—— (77)}$$

Subtract (77) from (74)

$$0 = (\sigma_1 - \sigma_2) [n^{(2)}]^T n^{(1)}$$

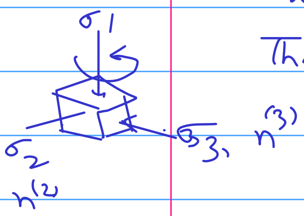
Assumption $\sigma_1 \neq \sigma_2$

$$\Rightarrow [n^{(2)}]^T n^{(1)} = 0 \Rightarrow \vec{n}^{(2)} \cdot n^{(1)} = 0 \Rightarrow n^{(1)} \perp n^{(2)}.$$

Similar arguments can be put for $n^{(2)}, n^{(3)}$, and also of $n^{(1)}, n^{(3)}$ pairs.

Special Cases of Roots of Eqn (69)

$$\sigma_1 > \sigma_2 = \sigma_3$$
$$n^{(1)} \quad n^{(2)} \quad n^{(3)}.$$



we have found $n^{(2)}$ & $n^{(3)}$ corresponding to σ_2 & σ_3
That are both are satisfying the Eqn

$$\tau \hat{n} = \sigma_2 \hat{n}.$$

Now,

Any vector $\hat{n}$ in the plane spanned by these two vectors can be written in the form

$$\hat{n} = c_2 n^{(2)} + c_3 n^{(3)} \quad \text{——} \quad (78)$$

$$\tau \hat{n} = \tau \left[c_2 n^{(2)} + c_3 n^{(3)} \right]$$

$$= c_2 \tau n^{(2)} + c_3 \tau n^{(3)}$$

$$= \sigma_2 (c_2 n^{(2)} + c_3 n^{(3)}$$

or $\tau \hat{n} = \sigma_2 \hat{n}$.

This proves that $\hat{n}$ is also an eigenvector of τ with eigenvalue σ_2.

Hence, All vectors in the plane spanned by $n^{(2)}$ and $n^{(3)}$ are principal stress directions associated σ_2.

Case - II $\sigma_1 = \sigma_2 = \sigma_3$.

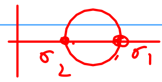
Same argument $\longrightarrow$ any unit Vector will be corresponding to σ_1

all planes will have normal traction of magnitude σ_1.

Hydrostatic Case

Maximum Minimum, Saddle Point. for normal Traction

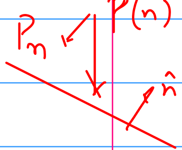
$$\frac{d\sigma}{d\theta} = 2\tau$$



in 3D $\to$ Locally Stationary Values.

* absolute Max
* absolute min.
* Saddle point in 3D space of unit Vector.

To prove this



Normal Component of Traction

$$P_n = \overline{P} \cdot \hat{n} = \hat{n} \cdot \vec{P}.$$

$$= [\hat{n}]^T P$$

$\Rightarrow P_n = n^T P$ But

$$\tilde{P} = \tau \hat{n}.$$

$$\vec{P}_n = \underbrace{\hat{n}^T \tau \hat{n}}. \quad \text{———} \quad (80).$$

Ex for 1 - $\hat{n} = \langle n_x, n_y, n_z \rangle$ $\quad \tau = \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}$

$$P_n = \tau_{xx} n_x^2 + \tau_{yy} n_y^2 + \tau_{zz} n_z^2 + 2\tau_{xy} n_x n_y + 2\tau_{xz} n_x n_z + 2\tau_{yz} n_y n_z \quad \text{--- (81)}$$

Interest $\rightarrow$ Extreme Value of P_n.

Keep in mind $\underbrace{n_x^2 + n_y^2 + n_z^2 = 1}_{\rightarrow \text{ Constrain.}}$

$$f(n_x, n_y, n_z) = \text{constant}.$$

Geometrically :- Variation of P_n on the Surface of unit sphere in the $n_x n_y n_z$ space

Theory of Lagrange's Multiplier

The constrained Max (or Min) will occur at a point at which

grad (P_n) is $\|^{el}$ to grad (f) — Constraint eqn.

Component of grad(P_n) — Variables n_x, n_y, n_z.

$$\frac{\partial P_n}{\partial n_x} = 2\underbrace{(\tau_{xx} n_x + \tau_{xy} n_y + \tau_{xz} n_z)}_{\rightarrow \text{ Nothing But } x\text{-component of } P(\vec{n})}.$$

$$= 2P_x$$

$$\text{grad}(P_n) = \left(\frac{\partial P_n}{\partial n_x}, \frac{\partial P_n}{\partial n_y}, \frac{\partial P_n}{\partial n_z}\right) = 2\vec{P} \quad \text{--- (83)}$$

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$$\text{grad } f = \text{grad}\,(n_x^2 + n_y^2 + n_z^2)$$
$$= 2(n_x + n_y + n_z)$$
$$= 2\hat{n}$$

If the two gradients are parallel

Then.

$$2\vec{P} \propto 2\hat{n}$$

or, $$2\vec{P} = \sigma\, 2\hat{n} \qquad \text{(84)}$$

or. $\underline{P = \sigma \hat{n}}$ — The condition for defining principal stress and their associated direction

$\Rightarrow$ Principal Directions also defines the planes on which normal traction takes on stationary values.

— x —